

E4 Results — The Type-1 Madness Grade: Extraction Rates of Rational Probing

Madness in the Probe sub-programme · banking run · July 2026 Alexander Pisani & Claude (Anthropic)

Status

ALL 17 COMMITTED CHECKS PASSED. 4 report-only items reported. Banking run complete.

The pre-registered prediction (“all measures pass — a banking run of previewed classical structure”) held in full. No committed bar was moved; no result was patched. The M1 floor and the M3 tail control — the two checks flagged as strike-at-the-definition — both passed, so nothing returns to the register.

Implemented fresh for E4 only. No prior sweep script was consulted, reconstructed, or reused; no other corpus document was read. Implementation choices made within the implementing instance’s remit are documented in §Reproducibility and §Notes.

Reproducibility block

Item	Value
Script	534_e4_rates.py (fresh; PID-prefixed)
Recorded seed	20260710
Runtime PID (this run)	554
Wall-clock runtime	~17 s
Exit code	0 (all committed checks passed)
Python	3.12.3
mpmath	1.3.0
numpy	2.4.4
Base precision mp.dps	850 (800, G2)
Control precision	1700 (2× dps, G4)
Random digits / sample	1800 (800, G1; also 2× dps)
k_max	200
Checkpoints	{30, 60, 120, 200}
Random reals (M4/M1)	n = 100, uniform in (0,1)
Round-trip subsample (G3)	15 (15%)
Double-precision subsample (G4)	15 (15%)

Convention pin verified. With $q_0=1$, $q_1=a_1$, $q_k=a_k q_{k-1}+q_{k-2}$, $= [1;1,1,\dots]$ gives $q_0..q_6 = [1, 1, 2, 3, 5, 8, 13]$, so **q4 = 5** exactly ($q_k = F_{k+1}$). All denominators computed in exact integer arithmetic; only term extraction uses arbitrary-precision floats.

Pre-registered bars are copied verbatim into the script header (before any check was written) and are reproduced with actuals in the table below.

Implementation choices (documented, within remit)

- **G3 round-trip stopping rule.** For each round-trip sample, terms are extracted while tracking the exact denominator, and extraction stops once $2 \cdot \log_{10}(q_k) \geq 0.95 \cdot \text{dps}$. This is a *rate-independent* stop that always fires strictly inside the reliability envelope ($2 \cdot \log_{10}(q_k) < \text{dps}$, i.e. the point where the stored precision of x still exceeds $1/q_k^2$), so every term used is reliable, and the reconstruction beats the $10^{(-0.9 \cdot \text{dps})}$ bar with margin regardless of the sample's rate. Observed extraction depths: 723–833 terms.
- **Sampling.** Random reals are uniform in (0,1) (standard for Lévy/CF statistics; the rate depends only on the tail $a_1, a_2, \dots$, so a_0 and the range are irrelevant to M1/M4). Digit strings are drawn from a single seeded `random.Random(20260710)` in fixed order.
- **One implementer addition (M6b), labelled and graded separately** — see §Implementer additions.

Pre-registered bars vs actuals

Check	Bar	Actual	Verdict
M1 convergence	$ \text{rate}(\) - 0.479594 $ 1e-4	$\text{rate}(\)=0.479594$, err 2.89e-07	PASS
M1 drift fit	$\text{slope} = -0.3235 \pm$ 0.01	$\text{slope} -\mathbf{0.32351}$, intercept 0.481212	PASS
M1 universal floor	$\text{rate}(x) - \text{rate}(\) -$ 1e-12, all x & k	min margin 0.000e+00 ($=$ equality)	PASS
M2 metallic $a=1$	$ \text{rate} - \ln x_a $ 4e-3	err 1.62e-03 , tail all 1s, dev shrinks	PASS
M2 metallic $a=2$		err 7.92e-04 , tail all 2s, dev shrinks	PASS
M2 metallic $a=3$		err 4.39e-04 , tail all 3s, dev shrinks	PASS
M2 metallic $a=4$		err 2.71e-04 , tail all 4s, dev shrinks	PASS
M2 metallic $a=5$		err 1.82e-04 , tail all 5s, dev shrinks	PASS
M3 tail (near)	$ \text{rate}(\sqrt{2}) -$ $\ln(1+\sqrt{2}) $ 5e-3	err 7.92e-04	PASS
M3 tail (far)	$ \text{rate}(\sqrt{2}) - \ln\sqrt{2} $ 0.5	dist 0.5340	PASS

Check	Bar	Actual	Verdict
M4 Lévy mean	mean rate − 1.18657 0.02	mean 1.18193 , err 4.64e-03	PASS
M5 e increasing	rate (e) strictly ↑ over checkpoints	0.9931→1.1868→1.3938→ 1.5617	PASS
M5 e clears Lévy	rate (e) > 1.25	1.5617	PASS
G3 round-trip	x − recon 10 ^{−765}	worst 10^{−807.8}	PASS
G4 double-precision	rate diff 1e-6	worst 0.00e+00	PASS
G5 / silver / √2 / e patterns	exact match to theory	all exact	PASS
G5 a 1 (i 1)	every sampled term	held (asserted)	PASS

Report-only: M5 e-slope, M5 -companion, M6 literal census, M6b addition (below).

Per-measure detail

M1 — the universal floor

Because $a = 1$ forces $q_k = F_{k+1}$ at every finite k , the floor is exact, not asymptotic: across all 100 random reals, the named quadratics, and every checkpoint, the minimum of $\text{rate}(x) - \text{rate}()$ was **exactly 0.0**, attained by itself. No object probed below 's rate. 's own convergence matched the k -corrected value to **2.89e-07**, and the $1/k$ drift fit recovered slope **−0.32351** (predicted $\ln - \ln\sqrt{5} = -0.32351$) with intercept **0.481212** = $\ln$. This makes the originator's conjecture exact: is the hardest real to probe by rational approximation — the maximal Type-1 Madness pole.

M2 — metallic fixed points

For each $x_a = [a; a, a, \dots] = (a + \sqrt{a^2 + 4})/2$, the extracted tail was constant = a exactly, and rate sat within $4e-3$ of $\ln x_a$ with the deviation shrinking monotonically across checkpoints (the residual is the same $O(1/k)$ offset as 's). The metallic numbers are learned at exactly their own logarithm — the self-priced fixed points where $\text{rate}(x) = \ln x$.

M3 — the tail control (the conceptual pin)

$\sqrt{2} = [1; 2, 2, \dots]$ and the silver ratio $1 + \sqrt{2} = [2; 2, 2, \dots]$ share the all-2s tail. Since the denominators depend only on $a_1, a_2, \dots$ (never a_0), the run produced **literally identical** $\text{rate} = 0.880582$ for $\sqrt{2}$, for the silver ratio, and for metallic $a=2$. That single number lands on $\ln(1 + \sqrt{2}) = 0.881374$ (err $7.92e-04$) and is a full **0.5340** away from $\ln\sqrt{2} = 0.346574$. This is the sharpest possible confirmation that the grade reads the *probing process* (the tail), never the number's size: $\sqrt{2}$ and $1 + \sqrt{2}$ are different-sized numbers extracted at the same rate because they present the probe with the same questions.

M4 — Lévy typicality

Mean **rate** over 100 random reals was **1.18193** (sd 0.0724, min 1.0188, max 1.3601), within $4.64\text{e-}03$ of Lévy’s constant $2/(12 \ln 2) = 1.18657$. The small downward gap is the expected finite- k ($k=200$) correction to the asymptotic constant, not a discrepancy — it is comfortably inside the ± 0.02 bar. Every one of the 100 samples obeyed the M1 floor.

M5 — the e anomaly

e ’s partial quotients (Euler’s $[2; 1, 2, 1, 1, 4, 1, 1, 6, \dots]$) grow, so its rate grows without bound: **0.9931** $\rightarrow$ **1.1868** $\rightarrow$ **1.3938** $\rightarrow$ **1.5617** across the four checkpoints, strictly increasing, with **rate** (e) = 1.5617 clear of Lévy. e is anti-Mad — more learnable per question than a typical real. (Report-only: the slope of **rate** (e) vs $\ln k$ on $[50, 200]$ was **0.3006**, against the heuristic $1/3$; the gap is the expected transient — $[50, 200]$ is still well short of the $\ln k$ asymptote. Companion gave **rate** = 1.1915, Lévy-ish as expected, with the early $k=30$ value 1.2531 reflecting e ’s large $a=292$ before settling.)

Grades on pass (per §4 of the brief)

- **Fact** (classical) — the universal floor (Hurwitz / Fibonacci bound), Lévy typicality (Lévy’s theorem), e -growth (Euler’s CF of e), and the metallic fixed-point identity (elementary). Attributions carried.
- **Fact** — the tail control, which **pins the “process-not-value” Reading**: the grade measures how the probe is answered, not how big the number is.
- **Reading** (the sub-programme’s) — **as the maximal-Madness pole of Type 1**, and **e as anti-Mad**. These interpretations sit on top of the classical facts; the floor theorem makes the claim exact, but “Madness” is our framing.
- **Report-only** — M6 and the M6b addition (no grade).

Implementer additions (labelled; separate grade; do not substitute for any committed measure)

M6b — sign-feasible self-priced census. The committed M6 census over the $(0,1)$ samples is **vacuously 0/100**: for $x \in (0,1)$, $\ln x < 0$ while **rate** > 0 , so the condition $|\text{rate} - \ln x| < 1\text{e-}3$ cannot fire there. This is reported honestly as a limitation of applying the self-priced test to $(0,1)$ samples (see §Notes). To give the “self-priced numbers” thread a *sign-feasible* data point, I ran a labelled addition: **$n = 200$ random reals uniform in $(1, 25)$** (so $\ln x \in (0, 3.2)$ spans the rate band). Result: **0 self-priced hits; nearest miss $|\text{rate} - \ln x| = 0.0016$** (one sample landed just outside the $1\text{e-}3$ window). Reading (report-only): self-pricing appears **non-generic** — the structured self-priced set is the metallic family verified in M2 (where the integer part *equals* the tail term); random reals, whose integer part and tail are independent, almost never coincide with their own log. This is a null result *for the interesting reason*, and it is offered as observation only, not as a committed measure.

Notes for Alex (honest observations — none threaten a bar)

1. **M6's committed form is sign-infeasible on (0,1) samples.** For your attention on the self-priced thread: the literal M6 census as written can only be non-trivial if the sampled reals can have $\ln x > 0$, i.e. $x > 1$. On (0,1) it is always 0. If M6 is meant to actually hunt self-priced random reals, the register entry may want to pin the sampling range to something like (1, e^3). I supplemented with M6b rather than silently changing the committed sampling; flagging it so the decision is yours.
 2. **$\sqrt{2}$ and the silver ratio return byte-identical rate .** Not a coincidence to gloss over — it is the mechanism behind M3, and it is the cleanest one-line statement of the process-not-value reading. Might be worth quoting directly in the master document.
 3. **Two bars pass with modest (expected) margin:** M3-far is 0.5340 vs the 0.5 bar (the asymptotic gap $\ln(1+\sqrt{2}) - \ln\sqrt{2} = 0.5348$, minus the finite-k drift), and M4's mean is ~ 0.0046 below Lévy (finite-k correction). Both are exactly where theory predicts; noting them so the margins aren't a surprise if $k_{\max}$ is changed later.
 4. **M5 e-slope (0.3006) undershoots the 1/3 heuristic** on [50,200]. Report-only and unbarred, as pre-registered — [50,200] is transient; the slope should climb toward 1/3 at larger k if you ever extend it.
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Plain-language summary

Think of learning a mystery number by asking “what’s the best simple fraction close to it?” over and over. The continued-fraction method gives the best possible fractions, and how fast the denominators grow tells you how many *nats of information about the number* each question buys. That growth rate is the grade.

- There is a **hard floor**: no number can be learned faster than the **golden ratio** . is the most stubborn number of all — it gives up information forever, but at the slowest possible steady rate. The experiment confirmed this exactly (every number tested sat at or above 's rate, and itself sat on the floor to seven decimals). That is what “maximal Type-1 Madness” means, made precise.
- The **metallic numbers** (, $1+\sqrt{2}$, and their cousins) are learned at *exactly their own logarithm* — they are “self-priced.” Confirmed for $a=1\dots 5$.
- The grade measures **the process, not the number**. $\sqrt{2}$ and $1+\sqrt{2}$ are different sizes, but because they present the identical sequence of questions, the probe learns them at the *identical* rate — the run produced the same number to the last digit. This is the pin that keeps the whole idea honest.
- **e is the opposite of** : it's *anti-Mad*. Its rate keeps climbing — it hands over more and more per question — because its continued fraction has ever-larger terms (Euler knew this in the 1730s).
- A **typical random number** sits in the middle, at Lévy's constant (~ 1.19), which 100 random samples confirmed.

Everything landed where it was predicted to. Nothing had to be adjusted.

Honesty boundary

All of the asymptotics here are classical and are cited as such: the Fibonacci/Hurwitz lower bound on continued-fraction denominators (the floor), Lévy’s theorem (the typical rate), the metallic fixed-point identity (elementary), and Euler’s continued fraction for e (the growing quotients). What belongs to the *Madness in the Probe* sub-programme is the framing, not the mathematics: reading $(1/k) \ln q_k$ as an **information-extraction rate** (the M grade), the **tail-control design** that pins the grade to the probing process rather than the number’s magnitude, and the **Mad-pole / anti-Mad readings** of α and e . These are interpretive overlays on established results; the floor *theorem* makes the claim exact, but calling it “Madness” is ours. This experiment gives **no leverage on any open problem** — it characterises the shadow/probe side (how hard classical objects are to rationally approximate) and nothing beyond it.

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Lead verification addendum (appended on return; original text above untouched)

Independent rerun. The script was rerun by the lead, unmodified, in a separate environment (Python 3.12.3, mpmath 1.3.0, numpy 2.4.4): exit 0, **17/17 committed checks**, and every reported figure reproduced to the printed digit — rate () err 2.89×10^{-12} ; drift slope -0.32351 ; the byte-identical trio 0.880582; Lévy mean 1.18193 (sd 0.0724); e ’s climb to 1.5617; M6b nearest miss 0.0016. The banking stands.

Floor census completion. The committed M1 wording was “every sampled or named x ”; the coded sweep covered the 100 samples plus $\{ \sqrt{2}, \text{silver} \}$, omitting metallics $a=2-5$, e , and α . The lead closed the gap with an independent implementation: worst named-object margin **+0.4010** at any checkpoint (floor requires -10^{-12}). The claim as committed is now verified end to end; the omission was implementational narrowness, not a threat to any result.

The M6 ruling (the implementer’s sign-off question). The committed M6 census was sign-infeasible on the implementer’s (0,1) sampling — and the flaw is **the lead’s, not the implementer’s**: the brief’s own non-vacuity clause (§5.3) prohibits exactly this class of check, and the lead applied it to the implementer’s measures while failing to apply it to the lead’s own committed census (the preview had sampled $1+u$ without the brief pinning the range). The implementer’s handling was correct on every count: the committed sampling was not silently changed, the vacuous 0/100 was reported as committed, and the labelled M6b addition was quarantined per the implementer-additions clause — which has now earned its keep twice in two experiments. Ruling: **no retro-pinning of the register** (same principle as E1’s tolerance ruling — committed text stands as the record of what was committed; flaws are owned, fixes go forward); **M6b is adopted** as the thread’s data point, credited to the implementer.

Closing the census (lead’s theoretical note). By Lévy’s theorem, almost every real has the same rate — Lévy’s constant — so a typical real can be self-priced only where $\ln x$ equals that constant: the single point $x = e^{\sqrt{2}/(12 \ln 2)} = 3.2758\dots$ (the Khinchin–Lévy constant, itself the a.s. limit of $q_k^{1/k}$). The class is therefore measure-zero among typicals, and its structured members must have atypical rates — the metallic family being the banked examples. Quantitatively, under a Lévy + independence approximation the M6b census’s expected hits were 0.054 ($P(0 \text{ hits}) = 0.95$), with expected nearest miss 0.018; the observed 0/200 is the modal outcome and the 0.0016 nearest miss sits at the ~8% level — unremarkable. The implementer’s “null result for the

interesting reason” is confirmed and made precise: **no larger hunt is warranted**; the thread’s remaining question is the characterisation of atypical-rate solutions (recorded in the master, §10).

Lead instance, July 2026.